

# How AI Answers

In Vector Space, you watched a token's final vector land in its neighborhood on the map. That landing was about meaning: the model understanding the words you put in. Now we move to the next step: how AI answers you.

Start with your phone. As you type a text, it suggests the next word: three choices above the keyboard, picked from the last word or two. And everyone gets pretty much the same suggestions.

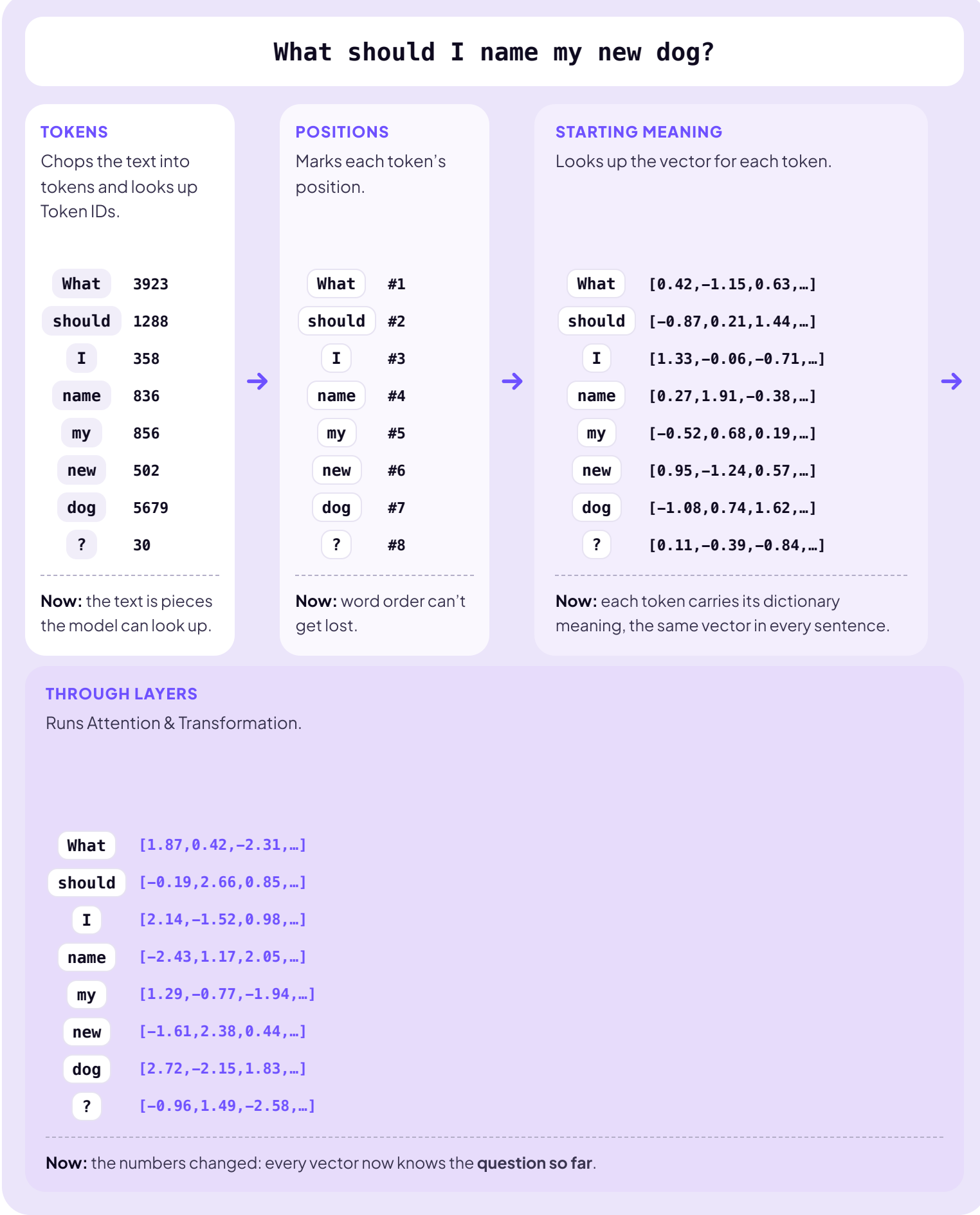
See you \_\_\_\_\_ .

soon

tomorrow

later

AI follows the same basic logic as your phone's suggestions, but, as you might expect, it goes **way deeper**. Let's take a simple question and watch AI build its answer.



## WHERE WE STAND

Look at what the model has now. The question isn't words anymore. It's eight rich vectors, each one carrying what its token means inside this exact question. **Meaning: established.**

Now, one detail changes everything: **the last token matters most**. Why? The model is about to write, and the next word goes in exactly one place: right after the last token. So that's the vector the model reads to pick it. And the layers have been preparing that vector all along: in every layer, attention folds the earlier tokens into the later ones, so **the last token's vector carries the entire question**.

By the final layer, the last token isn't really about the question mark (?) anymore. Its job has changed. Remember what the model practiced during training: predicting the next word, over and over, billions of times. **Training tuned the layers to turn what the text *means* into where the next word *lives***. So the vector carrying the question gets pushed across the map, and it lands in the neighborhood of the words most likely to come next: the first word of the answer.

## THE ANSWER, TOKEN BY TOKEN

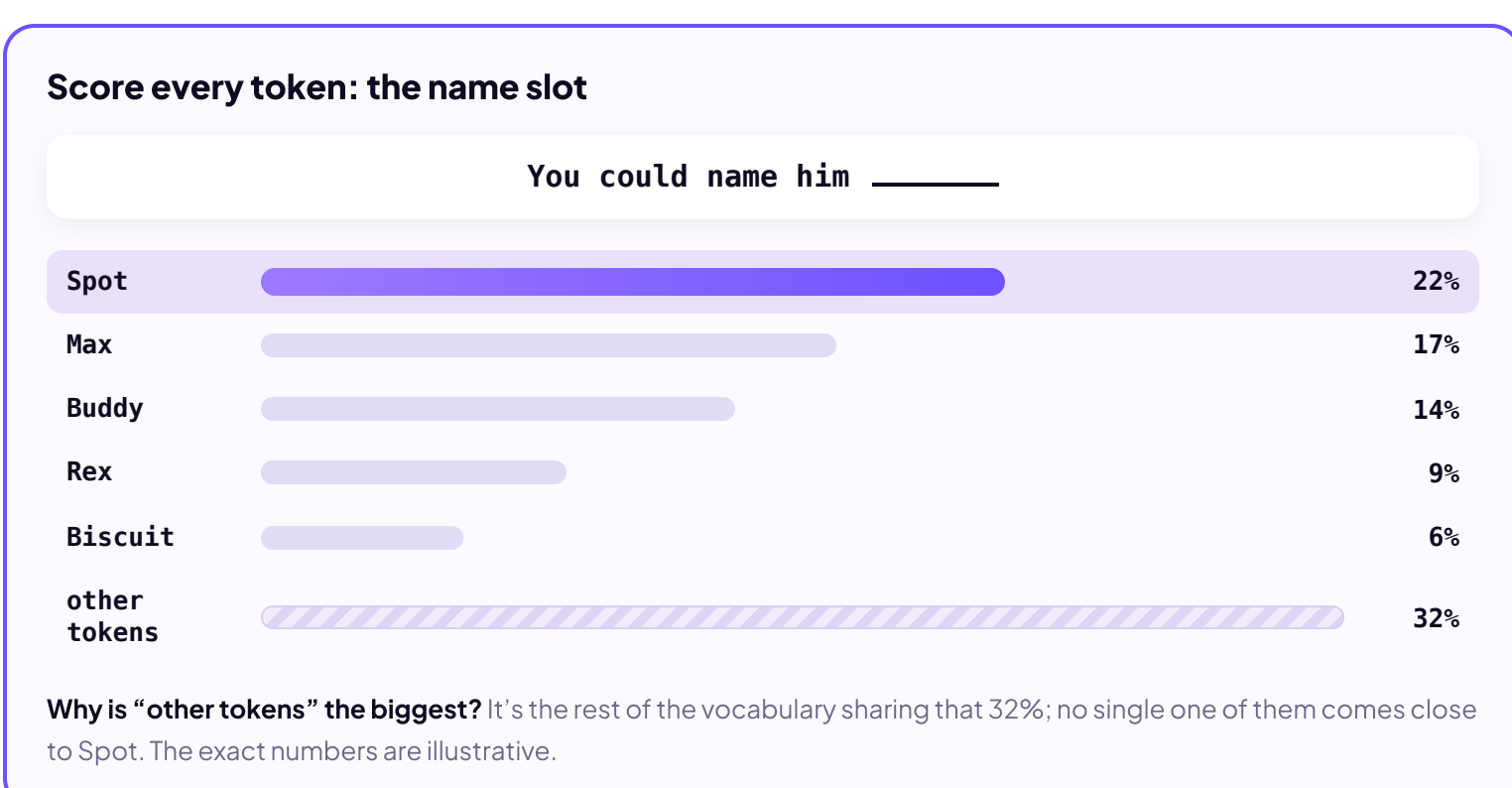
Now watch it happen. Each row below is one token of the answer, and in this example every token is a whole word. The model looks at the last token (the first column), checks which neighborhood its vector landed in, and takes the best-scoring word there. That word gets typed, joins the context, and becomes the new last token. Then the next row repeats the move. That move is called **prediction**.

| What should I name my new dog?                  |                        |                        |                                                               |
|-------------------------------------------------|------------------------|------------------------|---------------------------------------------------------------|
| LAST TOKEN                                      | REPLY SO FAR           | NEIGHBORHOOD           | TOP PREDICTIONS                                               |
| ?                                               | —                      | <i>Reply starters</i>  | <div>You 18%</div> <div>A 14%</div> <div>Great 9%</div>       |
| You                                             | You —                  | <i>Ways to suggest</i> | <div>could 24%</div> <div>can 12%</div> <div>should 10%</div> |
| could                                           | You could —            | <i>Naming verbs</i>    | <div>name 31%</div> <div>call 19%</div> <div>try 7%</div>     |
| name                                            | You could name —       | <i>Who gets named</i>  | <div>him 45%</div> <div>your 21%</div> <div>the 8%</div>      |
| him                                             | You could name him ... | <i>Dog names</i>       | <div>Spot 22%</div> <div>Max 17%</div> <div>Buddy 14%</div>   |
| <div><div>✓</div>You could name him Spot.</div> |                        |                        |                                                               |

You asked for a name for your new dog, but the model didn't start with a name. It started with You, the most likely first word of a reply. Read the Neighborhood column from top to bottom and you can watch the sentence close in: reply starters, ways to suggest, naming verbs, who gets named, and only then dog names. The name arrived when the sentence had built a slot that only a name could fill. Five rows, one move, and a context that grew by one word each time.

## THE RANKED LIST

One more idea to clarify. The table shows three predictions per row, but that's just the top of the list. In reality, the model assigns a probability to every token in its vocabulary, hundreds of thousands of them, and ranks them all. That's what "landing in a neighborhood" really is: the neighborhood is the top of the ranking. Here's the name slot under the microscope: the moment Spot arrived.



The model takes the top of the list and types it: Spot.

And that's the whole machine. Everything you just watched, from tokens to vectors to a ranked list to the next word, has a name: **inference**. It's what the model does every time it answers you.

Every answer is built one token at a time.

The whole run is called inference.